Experiment 9: Hive Operations - CREATE, SHOW, DESCRIBE (Windows)

Aim

To download, install, and configure Apache Hive on a Windows operating system and implement

fundamental Data Definition Language (DDL) operations: CREATE, SHOW, and DESCRIBE. Procedure
1. Download a stable binary release archive of Apache Hive and extract the package directories

directly into a clean root path (e.g., C:\hive). 2. Map your system path routing dependencies by creating a new HIVE_HOME system variable pointing
to C:\hive and adding %HIVE_HOME%\bin to your environment path. 3. Launch your core background Apache Hadoop cluster processes (start-dfs.cmd and start-
yarn.cmd) from an elevated Administrator Command Prompt window. 4. Execute HDFS commands to build storage layer directory paths (/tmp and /user/hive/warehouse)
and configure loose group write permissions using chmod flags. 5. Initialize the Hive local embedded metastore schema directory structural mappings by running the
schematool -dbType derby -initSchema command utility. 6. Launch the interactive Apache Hive console interface client directly inside your Command Prompt
window by typing the hive runtime command. 7. Run operational commands to create a structured relational table schema layout with defined data

types, list the available tables, and inspect column metadata configurations. Commands

Windows Console Configuration Commands
cmd
:: 1. Ensure your core background Hadoop storage daemons are active
jps

:: 2. Initialize the local Hive metadata schema using the Derby database engine cd C:\hive\bin
schematool -dbType derby -initSchema

:: 3. Launch the interactive Hive terminal client console shell
hive

Use code with caution. Interactive Hive Shell Queries
sql

- - 4. Create a structured relational table schema to process employee recordsCREATE
TABLE IF NOT EXISTS default.employee_records (

emp_id INT,

emp_name STRING,

salary FLOAT,

department STRING

)ROW FORMAT DELIMITED

FIELDS TERMINATED BY ',';

- - 5. List and display all tables registered in the current database workspaceSHOW
TABLES;

- - 6. Inspect the structural design column definitions of your table schemaDESCRIBE
employee_records;

- - 7. Exit the interactive Hive terminal client safely

EXIT;

Use code with caution. Execution & Output

text

C:\hive\bin>hive

hive> CREATE TABLE IF NOT EXISTS default.employee_records (

> emp_id INT,

> emp_name STRING,

> salary FLOAT,

> department STRING

>)

> ROW FORMAT DELIMITED

> FIELDS TERMINATED BY ',';

OK

Time taken: 0.45 seconds

hive> SHOW TABLES;

OK

tab_name

employee_records

Time taken: 0.12 seconds

hive> DESCRIBE employee_records;
OK
col_name data_type comment
emp_id int
emp_name string
salary float
department string
Time taken: 0.18 seconds